



Fabio Slika Stella
Embedded Software Engineer
Embedded Linux & IoT • C++/C Developer

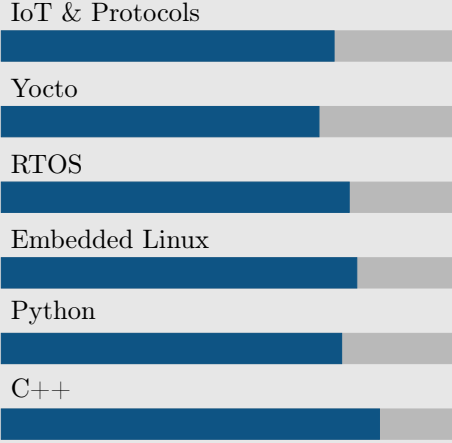
- 30 years old
- Brazil – Remote
- +55 (46) 9 9106-8741
- <https://github.com/Gedankenn>
- fabioslikastella@gmail.com

About Me

Computer Engineer with hands-on experience in low-level software, Embedded Linux and real-time C++/C firmware. Career across access control, railway, automotive and industrial automation. Comfortable with Linux kernel and device-driver concepts, Yocto builds, RTOS and IoT communication protocols, with a consistent focus on reliability, automation and debugging close to the hardware.

Currently working on Embedded Linux and firmware for IoT access-control devices at ControlID – Yocto, bootloaders, OTA and C++ across TI and ST MCUs. Open Source side projects under stellacorp.com.br.

Skills



(*)[Scale 0 (Basic) a 6 (Advanced)]

Career Objective

Senior-level role in embedded software and IoT development, applying modern C++ and Python, Embedded Linux and real-time systems to architect reliable, maintainable connected devices.

Education

2014–2022 Bachelor's Degree in Computer Engineering Pato Branco
Federal University of Technology – Paraná (UTFPR)

Languages

Portuguese Native
English Advanced

Professional Experience

2026–Present	Embedded Software Developer II	ControlID
	Embedded Linux and firmware for IoT access-control devices: Yocto builds, bootloaders and OTA pipelines, C++/C development across TI and ST MCUs, communication protocols, diagnostics and debugging.	
2024–2025	Embedded Systems Developer	Progress Rail – Caterpillar
	Low-level C/C++ firmware for safety-critical railway modules; system validation under operational and stress conditions; real-time constraints and industrial communication protocols.	
2024	Software Engineer (Embedded)	Zeentech / Stellantis
	Embedded C/C++ modules for automotive systems; support in integration and validation; low-level timing and stability investigations.	
2022–2024	Embedded Systems Developer	InoBram
	STM32 firmware and custom bootloader maintenance; UART, I2C and RS-485 protocol implementation; refactoring for stability and testability; in-circuit debugging.	
2022	Test Engineer	IAV Brazil
	Automotive embedded validation with Typhoon HIL and black-box testing; failure documentation and root-cause analysis support.	
2021–2022	R&D Intern	Xpert Automação
	Embedded firmware (ESP32) for fuel station automation; Linux-based systems, networking and OpenWRT integration.	
2019–2022	Researcher	UTFPR – Pato Branco
	Modeling and simulation of control algorithms for electric motors; performance analysis and validation tools.	

Technical Qualifications

- Modern C++ (C++17/20) with object-oriented design; C and Python for tooling and automation.
- Embedded Linux: Yocto builds, system bring-up, OTA pipelines and bootloaders.
- RTOS and real-time firmware (FreeRTOS, bare-metal) on STM32, ESP32, TI and ST MCUs.
- IoT and field protocols: MQTT, HTTP/IP, CAN/J1939, RS-485, I²C, SPI and UART.
- Linux-friendly tooling: Git, CMake, Docker, CI/CD and Shell/Python automation.
- Cross-platform / cross-development mindset with diagnostic rigor (GDB, JTAG/SWD, serial tracing).
- Kernel-adjacent foundation: memory, timing, interrupts, device-driver and boot flow concepts.
- Hands-on hardware bring-up and firmware–hardware integration with Ki-Cad.

Additional Courses

2023	Embedded Systems Bare-Metal Programming (STM32)	Udemy
	Online	
2023	IoT Application Development with ESP32 (ESP-IDF)	Udemy
	Online	
2020	Networking Academy Learn-A-Thon	Cisco
	Online	
2020	Introduction to Cybersecurity	Cisco
	Online	